MORGAN HUBERTY

Baltimore, MD | mhubert3@jh.edu | **U.S. Citizen**
linkedin.com/in/morganhuberty | github.com/mhubert3 | mhubert3.github.io

EDUCATION

Johns Hopkins University — Baltimore, MD
Bachelor of Science in Computer Engineering — *Expected May 2027*
Relevant Coursework: FPGA Lab, Computer System Fundamentals, Signals & Systems, Mastering Electronics, Data Structures
Honors: Childhood Cancer Canada STEM Biomedical Survivor Scholarship (2024)

TECHNICAL SKILLS

Hardware Design: VHDL, Verilog; FSM design, two-flop synchronizers, UART & I²C protocols, open-drain I/O, Xilinx IOBUF
Verification & EDA: Self-checking testbenches, assertions, Verilator (C++ drivers), Xilinx Vivado (synthesis, XDC), LTSpice
Software & Systems: C, C++, Python (NumPy, Pandas), MATLAB, Bash, Git, Linux, Docker; basic x86-64 assembly
Currently Learning: SystemVerilog, SystemC

HARDWARE DESIGN PROJECTS

ZipCPU Soft-Core Processor Build | *Verilog*, *Verilator*, *C++*, *Artix-7* — In Progress

- Working through Dan Gisselquist's ZipCPU tutorial curriculum toward building the ZipCPU, a 32-bit pipelined RISC soft-core processor, on a Digilent Cmod A7-35T (Xilinx Artix-7)
- Writing Verilog RTL and simulating each design with Verilator and C++ test drivers before synthesizing and validating it on hardware; formal verification with SymbiYosys is the next milestone

UART Transceiver | *VHDL*, *Xilinx Vivado*, *Cmod A7-35T (Artix-7)* — github.com/mhubert3/FPGA-Serial-UART

- Designed a parameterized UART transmitter and receiver (8N1, 115,200 baud from a 12 MHz clock) with FSM control, a two-flop synchronizer on the asynchronous RX input, and mid-bit sampling via a 1.5-bit-period start delay
- Verified with a self-checking closed-loop testbench using VHDL assertions on a worst-case alternating pattern, then validated on hardware through PMOD loopback and a MATLAB serial host interface

I²C Master Controller | *VHDL*, *Xilinx Vivado*, *Cmod A7-35T (Artix-7)* — github.com/mhubert3/FPGA-Serial-I2C

- Designed an FSM-based I²C master handling start, repeated start, stop, read/write, ACK/NACK, and clock stretching, with cycle-counted 5 µs bus timing and Xilinx IOBUF primitives for open-drain SDA/SCL
- Modeled bus pull-ups as weak-high signals in simulation; validated end-to-end on hardware by relaying bytes from a MATLAB GUI over UART to the FPGA, across I²C to a PIC16F18326 slave, and back

EXPERIENCE

Morgate Solutions | *Software Engineering Intern* — Summers 2024, 2026

Edge Software Optimization — *Jun 2026 – Aug 2026*

- Containerized C/C++ parsers with Docker for Linux edge gateways, optimizing memory use to fit 512 MB device limits
- Emulated network jitter and profiled CPU usage to find and prevent buffer overflows in high-frequency data streams

Low-Level Data & Protocol Bridging — *Jun 2024 – Aug 2024*

- Developed C extraction tools using struct packing and bitwise operations to decode 50+ GB of binary telemetry
- Managed struct byte alignment and endianness; built a Python test harness to validate parser output

COMPUTER ARCHITECTURE PROJECT

Cache Simulator | *C*, *C++* — github.com/mhubert3/Cache-Simulator

- Built a configurable set-associative cache simulator (sets, blocks, block size, LRU/FIFO replacement, write policies) that reports hit/miss counts and cycle totals from memory traces, used to compare associativity and policy trade-offs

LEADERSHIP

National Society of Black Engineers, JHU Chapter | *Co-President; former Vice President & Finance Chair* — Sep 2023 – Present

- Secured sponsorships from Medtronic, Visa, and Jane Street; hosted events with Medtronic and Shift Alt Her
- Expanded reach through events with JHU's Life Design Lab and Pava Center, collaborations with five student organizations including SHPE and BMES, cross-department and Instagram outreach, and a new alumni LinkedIn group
- Delegate committee work, run fundraisers and logistics, and fund member travel to regional and national conferences
- As Finance Chair, managed a $9K annual budget and secured $3.5K through fundraising and corporate sponsorships

HopThon, JHU Dance Marathon | *Fundraising Chair* — Fall 2025 – Spring 2026
Alpha Epsilon Pi | *Philanthropy Chair* — Spring 2025